



Compound Astragalus and *Salvia miltiorrhiza* extracts modulate MAPK-regulated TGF- β /Smad signaling in hepatocellular carcinoma by multi-target mechanism



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ABSTRACT

Ethnopharmacological relevance: *Astragalus membranaceus* Bunge (Leguminosae) and *Salvia miltiorrhiza* Bunge (Lamiaceae) are two important Chinese herbs with a long history of extensive ethnobotanical usage in the treatment of liver-related diseases over many centuries. Presently, these two herbs are being used either as a single herbal formulation or a composite formula for the treatment of liver related conditions. In response, recent studies on these two herbs have focused on elucidating their mechanisms of action, particularly with regards to their anti-hepatocarcinogenic effects. Previously, we have reported that Compound Astragalus and *Salvia miltiorrhiza* extract (CASE), a synergized composite extract from *Astragalus membranaceus* and *Salvia miltiorrhiza* ameliorates liver fibrosis and hepatocellular carcinoma (HCC) by modulating the TGF- β /Smad pathway. Meanwhile, MAPK activation and MAPK-dependent linker phosphorylation of Smad2/3 and their preferential nuclear import are crucial for overall oncogenic role of TGF- β /Smad signaling in HCC. To elucidate further, we studied the effect of CASE on the MAPK pathway and how it affects MAPK-dependent regulation of TGF- β /Smad signaling using both cell and animal models of HCC.

Materials and methods: We used immunofluorescence and western blot techniques to monitor effect of CASE on the activation of the MAPKs (pERK, pJNK and pp38) in TGF- β_1 -stimulated hepatic stellate cells (HSCs), HepG2 cells and also diethylnitrosamine (DEN)-induced HCC in rats. Also phosphorylation and subcellular distribution of pSmad2/3, Smad4 and Imp7/8 in TGF- β_1 -stimulated HSC and HepG2 cells were monitored. The expression of pERK, pJNK, pp38 and PAI-1 gene were monitored by using western blot technique. The effect of CASE on domain-specific phosphorylation of Smad2/3 and their subcellular distribution, and the expression of Smad4 and its subcellular distribution in TGF- β_1 -stimulated HSCs and HepG2 cells were evaluated by using immunofluorescence technique. And the expression of Imp7/8 and their subcellular distribution were assessed by both immunofluorescence and western blot techniques, while PAI-1 gene expression was assessed by western blot.

Results: *In vitro*, CASE in a concentration-dependent manner increased the expression of pp38 but decreased the expression of pERK and pJNK; however, *in vivo*, CASE in a dose dependent manner decreased the expression of pERK, pJNK as well as pp38. Also, CASE concentration dependently inhibited pSmad2C/L, pSmad3L, Smad4, Imp7/8 and their nuclear import; it had no effect on pSmad3C in HepG2 cells; significantly decreased PAI-1 gene expression in both *in vitro* and *in vivo*.

Conclusions: CASE blocked MAPK activation, MAPK-dependent linker phosphorylation of Smad2/3, Smad4 expression, Imp7 expression and their nuclear import leading to significant down-regulation of PAI-1 gene expression; further highlighting the multi-target anti-HCC effect of CASE and its potential drug candidature.

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1. Introduction

Astragalus membranaceus Bunge (Leguminosae) and *Salvia miltiorrhiza* Bunge (Lamiaceae) are two Chinese herbs with a long recorded history of use as a hepatoprotective medicine spanning many centuries (Roxas and Jurenka, 2007). As a result, the past decades have seen increased scientific investigations on these two herbs, particularly their efficacy and safety in cell and animal models of diseases including liver fibrosis and hepatocellular carcinoma (HCC) (Chan et al., 2009; Chen et al., 2011; Roxas and Jurenka, 2007; Zhang et al., 2013) as well as some bioactivity and phytochemical profile of some of their active components (Cho and Leung, 2007a, b; Shao et al., 2004; Shen et al., 2006). Our study group (Yang et al., 2008) together with others (Jun, 2004; Lee et al., 2006) designed a synergized formula comprising astragaloside, astragalus polysaccharide and salvianolic acid, the active compounds from *Astragalus membranaceus* and *Salvia miltiorrhiza* respectively, known as Compound Astragalus and *Salvia miltiorrhiza* extract (CASE) using orthogonal studies. At the mechanistic level, CASE was shown in both *in vitro* and *in vivo* studies to modulate the TGF- β /Smad signaling pathway to inhibit TGF- β -specific target gene expression in liver fibrosis and HCC and these effects led to amelioration of HCC phenotypic hallmarks (Cell proliferation, cell migration and invasion) (Yang et al., 2008; Liu et al., 2010; Rui et al., 2012; Rui et al., 2014; Hu et al., 2014). Meanwhile the overall oncogenic signaling output of dysregulated TGF- β /Smad in HCC is substantially augmented by the mitogen activated protein kinase (MAPK) pathway (Giehl et al., 2007; Zhang, 2009) through linker-specific phosphorylation of Smad2/3 and their preferential nuclear relocation (Fuentelba et al., 2007; Hata and Davis, 2009; Kretschmar et al., 1999). This MAPK-dependent linker phosphorylation of Smad2/3 and their subsequent nuclear import are crucial for MAPK-regulated TGF- β /Smad signaling in HCC (Hayashida et al., 2003; Yoshida et al., 2014). In an earlier study by our group, CASE was shown to inhibit TGF- β ₁-induced activation of JNK and JNK-dependent linker phosphorylation of Smad2/3 in myofibroblast (Yang et al., 2008). On the basis of these previous results regarding JNK and our recent studies on MAPK-Specific inhibitors (PD98059 [ERK-Specific inhibitor], SP600125 [JNK-Specific inhibitor] and SB203580 [p38-Specific inhibitor]) which modulated the TGF- β /Smad signaling in HCC (Boye et al., 2015) we strongly suspect that CASE may target the entire MAPK pathway in a manner similar to the MAPK-Specific inhibitors to abrogate TGF- β /Smad signaling.

Consequently, we hypothesize that CASE may modulate the entire MAPK (ERK, JNK and p38) pathway, particularly MAPK activation and MAPK-dependent linker phosphorylation of Smad2/3 and their nuclear import in order to truncate the MAPK-regulated TGF- β /Smad signaling in HCC. To test the above hypothesis we studied the effect of CASE on the MAPK-regulated TGF- β /Smad signaling using both *in vitro* (HSC and HepG2 cells) and *in vivo* (DEN-induced HCC in rats) models of HCC.

2. Materials and methods

2.1. Preparation of astragaloside, astragalus polysaccharide and CASE

The herbs of *Astragalus membranaceus* Bunge (Leguminosae) and *Salvia miltiorrhiza* Bunge (Lamiaceae) were purchased from Bozhou Crude Drug Market (Anhui province, China) and authenticated by Professor Xiaoxiang Zhang (Department of Pharmaceutical Engineering of Hefei University of Technology) who is a specialist in traditional Chinese herbal medicine. Voucher specimens were deposited at the Traditional Chinese medicine specimen room (Anhui University of Chinese Traditional Medicine, Hefei, Anhui Province, China). The process of extracting and preparing the three components of

CASE were done as previously described (Yang et al., 2008; Liu et al., 2010). Briefly, they are described below.

2.1.1. Preparation of astragaloside and astragalus polysaccharide

A 10 kg quantity of chopped dried roots of *Astragalus membranaceus* were extracted by using 90% ethanol three times at 3 h in each case, followed by drying under low pressure. A yield of 1.75 kg of powdered extract obtained was dissolved in 17 L of water, filtered and the aqueous portion chromatographed on polystyrene resin (D101, 0.3–1.25 mm; Nankai Chemical Factory, Tianjin, China). Further, the aqueous portion was sequentially eluted with water, 40 and 70% ethanol, followed by drying under low pressure, finally yielding 33 g of dry powder. By using the colorimetric method previously described by Tang (2004), the purity of the astragaloside was estimated at 67.4%.

To prepare astragalus polysaccharide, the residue from the three times 90% ethanol extraction of the chopped roots of *Astragalus membranaceus* was twice decocted with 10 L of water for 1 h each time. After filtration of the decoction, the filtrate was concentrated to 5 L in a vacuum desiccator at 70 °C. Astragalus polysaccharide was precipitated by using 90% ethanol followed by dissolution in water. Again, 90% ethanol was added to the astragalus polysaccharide solution followed by retrieval of the astragalus polysaccharide by the method previously described (Wu et al., 2001). Subsequently, the resulting sediment was washed twice by using 80% ethanol, followed by drying under low pressure yielding 231 g dry powder. By using the phenol–sulfuric method as previously described (Zhang et al., 2001), the purity of astragalus polysaccharide was estimated at 55.4%. Both astragaloside and astragalus polysaccharides were stored at –80 °C until use.

2.1.2. Preparation of salvianolic acid

The extraction of salvianolic acid from *Salvia miltiorrhizae* was done by following the method previously described by Lee et al. (2006). Concisely, a 400 g of the dried roots of *Salvia miltiorrhizae* was powdered and extracted with 1 L of distilled water at 80 °C for 2 h. The resulting infusion was filtered and lyophilized (Virtis, freeze – mobile, NY) yielding 100 g of light brownish dry powder. By using a colorimetric method previously described by Ye (2006) the purity of salvianolic acid was estimated at 48.32%. The extract was stored at –80 °C until use.

2.1.3. Preparation of CASE

CASE was prepared by following the method previously described (Yang et al., 2008). Briefly, the powdered forms of astragalosides, astragalus polysaccharide and salvianolic acids were dissolved in 0.5% sodium carboxymethylcellulose (CMC-Na) according to a standard ratio (70:1:1.85) in weight of crude herbs.

2.2. Animal model of HCC and CASE treatment

Matured and healthy male Sprague–Dawley rats of body weight (180–200 g) were purchased from Xipuer-bikai Company (Shanghai, China). The rats were housed in conventional cages at 20–22 °C, supplied with standard laboratory chow and water *ad libitum*, and kept at a 12 h light/dark cycle. The rats were maintained under these conditions for at least 1-week for acclimatization before the commencement of experiments. The handling and use of the rats in the study were carried out in accordance with the guidelines for the humane treatment of animals set out by the Association of Laboratory Animal Sciences and the Center for Laboratory Animal Sciences at the Anhui Medical University. The rats were randomly divided into five groups of 10 rats each: the control group, the DEN treatment group and three CASE treatment groups. The rats in the control group were given normal animal chow, water and 0.5% CMC-Na by gavage; the rats in the DEN group in addition to daily animal chow and water

were treated with 0.2% DEN dissolved in 0.5% CMC-Na in the morning by gavage 5 times a week for 12 or 16 weeks to induce HCC; and the rats in the CASE groups, in addition to daily normal animal chow and water were concurrently treated with 0.2% DEN in 0.5% CMC-Na in the morning by gavage five times a week and CASE (60, 120 or 240 mg/kg respectively) in the afternoon by gavage per day for 12 or 16 weeks. The rats were sacrificed two weeks post-DEN treatment (12th or 16th-week). One lobe of liver from each rat in each group was harvested and stored at a temperature of -80°C until use.

2.3. Cell origin, culture and treatment of cells (hepatic stellate cells [HSCs] and HepG2 cells) with CASE

The use of animals in this study was approved by the Animal Ethics Committee of Anhui Medical University. HSCs were isolated from the liver of normal male Sprague–Dawley rats (450–500 g) by using collagenase and sequential Pronase-E digestion as previously described (Date et al., 2000). The isolated HSCs were > 90% pure (As determined microscopically based on vitamin A droplet-dependent auto-fluorescence) and had > 99% viability (Trypan blue exclusion test). The human hepatocellular carcinoma HepG2 cell line was purchased from The Chinese Academy of Sciences Cell Bank (Shanghai, China). The HSCs and HepG2 Cells were grown as sub-confluent monolayer cultures in Dulbecco's modified Eagle's medium (DMEM; Gibco, Rockville, MD, U.S.A.) supplemented with 10% fetal bovine serum (FBS, Sijiqing, Zhejiang Tianhang Biological Technology Co., LTD; Zhejiang, China), and maintained in a humidified 5% CO_2 incubator at 37°C . The experiments were performed at the log phase of growth after the cells had been plated for 24 h. HSC and/or HepG2 cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 mg/ml), and subsequently treated with TGF- β_1 (9 pmol/L) for 1 h. The cells of the control group were treated with an equal volume of serum-free medium.

2.4. Western blot analysis

To detect the effect of CASE on the expression of PAI-1, Importin7/8 and phosphorylation of MAPKs (ERK, JNK and p38), HSC and/or HepG2 cells were seeded at a density of 1×10^6 cells/25 cm^2 culture flasks, and then treated under the indicated conditions. Total proteins from HSC and/or HepG2 cells and frozen liver tissue were extracted by using Western blot and IP cell lysis liquid (Beyotime, Shanghai, China) as previously described (Wu et al., 2014). Proteins were separated by sodium dodecyl sulfate/polyacrylamide gel electrophoresis (SDS/PAGE), transferred onto polyvinylidene difluoride (PVDF) membranes (Millipore, Bedford, MA, USA) by wet transfer method, blocked in 5% skim milk powder dissolved in Tris-buffered saline solution/0.1 Tween 20 (TBST), incubated with the primary antibody overnight at 4°C , washed 3 times with TBST for 10 min each time, incubated with corresponding secondary antibody for 2 h at room temperature, washed 3 times with TBST for 10 min each time, and finally the membranes were developed by using the ECL chemiluminescence system (Amersham, Piscataway, NJ, USA). Primary antibodies used in this study included plasminogen activator inhibitor 1 (PAI-1, rabbit anti-PAI-1 antibody) (Santa Cruz Biotechnology, Santa Cruz, CA, USA); phospho-ERK1/2 and ERK1/2, phospho-JNK1/2 and JNK, phospho-p38 and p38 (rabbit anti-phospho-ERK1/2, anti-ERK1/2, anti-phospho-JNK1/2, anti-JNK1/2, anti-phospho-p38 and anti-p38 kinase antibodies) (Cell Signaling Technology, Beverly, MA, USA); Importin7 and 8 (Imp7/8, rabbit anti-Importin7 and anti-Importin8 antibodies) (Abcam, Cambridge, UK) and glyceraldehyde phosphate dehydrogenase (GAPDH) (mouse anti-GAPDH) (Cell Signaling Technology, Beverly, MA, USA). Densitometric analysis was carried out by using Quantity One software (Bio-Rad, California, USA).

2.5. Immunofluorescence analysis

To detect the effects of CASE on intracellular localization of Smads and Importin7/8, HSCs and/or HepG2 cells were seeded on slides in a 24-well plate and then treated under the indicated conditions. The cells were fixed with 4% paraformaldehyde for 30 min, permeabilized with 0.1% saponin for 10 min, and blocked with 0.5% bovine serum albumin in phosphate buffer saline (PBS) for 30 min at 4°C ; then incubated with each primary antibody overnight at 4°C , washed 3 times with PBS for 5 min each time, incubated with corresponding fluorescein isothiocyanate (FITC)-conjugated secondary antibody for 2 h at room temperature, washed 3 times with PBS for 5 min each time, incubated with 4',6-diamidino-2-phenylindole (DAPI; Sigma) for 10 min at room temperature for nuclear staining. Finally, slides were mounted with 80% phosphoglycerol, viewed and photographed under a fluorescence microscope (Olympus, Tokyo, Japan). Primary antibodies used in this experiment included Smad2/3 phosphorylated at the C-terminal region (rabbit anti-pSmad2C and pSmad3C antibodies) (Cell Signaling Technology, Beverly, MA, USA), Smad2/3 phosphorylated at the link region (rabbit anti-pSmad2L and pSmad3L antibodies) (A gift from Prof. K. Matsuzaki, Kansai Medical University, Japan), Smad4 (mouse anti-Smad4 antibody) (Santa Cruz Biotechnology, Santa Cruz, CA, U.S.A.) and Importin7 and 8 (Imp7/8, rabbit anti-Importin7 and anti-Importin8 antibodies) (Abcam, Cambridge, UK). At least 100 stained cells were analyzed per sample in each experiment.

2.6. Statistical analyses

Data were expressed as mean $\pm$ standard deviation (SD). Statistical analyses were performed by SPSS 11.0 for Windows (SPSS, Inc., Chicago, IL, U.S.A.). Experimental and control groups were compared by one-way ANOVA. $P < 0.05$ was considered statistically significant.

2.7. Theory/calculations

Transforming growth factor beta (TGF- β) is a multi-functional and a ubiquitous cytokine crucial in all cellular, developmental and homeostatic processes as well as disease pathogenesis. It employs basically two major signaling modes (Canonical and Non-canonical) in almost all the cellular processes it partakes. While the former signaling mode involves mediation by Smad proteins, the latter involves crosstalk with other signaling pathways, of which mitogen activated protein kinase (MAPK) pathway is integral. Many therapeutic modalities including but not limited to TGF- β -Receptor inhibitors have been designed to abrogate dysregulated canonical TGF- β signaling in diseases such as HCC but so far they have proved comparatively ineffective due to the potential of TGF- β to reactivate its non-canonical compensatory pathways mainly regulated in part by the MAPKs (ERK, JNK and p38). As a result the MAPK pathway has become a possible target for investigations to prospect for new targets for therapy, since it regulates the oncogenic arm of the TGF- β signaling in cancer. To this end we designed this study to investigate the effect of CASE, a potential anti-HCC herbal drug on the MAPK-regulated TGF- β /Smad pathway, on the basis of our previous finding on CASE regarding the TGF- β /Smad pathway where CASE ameliorated liver fibrosis and HCC progression by modulating the TGF- β /Smad pathway.

It is envisioned that the future success of CASE as a potential anti-HCC candidate drug will not only justify its long uneventful folk use in China but also provide a much cheaper and readily available alternative to conventional HCC drug therapies.

3. Results

3.1. CASE down-regulated TGF- β_1 -induced activation of pERK and pJNK but up-regulated pp38 in both HSCs and HepG2 cells

TGF- β_1 induced activation of ERK1 but not ERK2 in HSCs when compared to control, however, CASE (20, 40 and 80 $\mu\text{g/ml}$ respectively) concentration dependently decreased TGF- β_1 -induced up-regulation of ERK1 while at the same time completely abolished the activation of ERK2 (Fig. 1(1)). Similarly, TGF- β_1 induced activation of JNK1/2 in HSCs compared to control; however, prior treatment of HSCs with CASE (20, 40 and 80 $\mu\text{g/ml}$ respectively) concentration dependently decreased the hitherto increased TGF- β_1 -induced activation of JNK1/2 (Fig. 1(2)). TGF- β_1 stimulation of HSCs produced increased activation of pp38 compared to control, but interestingly, CASE potentiated TGF- β_1 -induced activation of pp38 in HSCs (Fig. 1(3)). Similarly, TGF- β_1 induced the expression of pERK, pJNK and pp38 in HepG2 cells after incubation of HepG2 cells with exogenous TGF- β_1 . However, pretreatment of HepG2

cells with CASE (20, 40 and 80 $\mu\text{g/ml}$ respectively) before stimulation of HepG2 cells with TGF- β_1 showed a concentration-dependent inhibition of the activation and expression of pERK (Fig. 1(4)). With respect to pJNK, TGF- β_1 stimulation produced an increase in pJNK activation compared to control group. Though prior CASE treatment of HepG2 cells before TGF- β_1 stimulation inhibited pJNK activation and expression, it was only significant at a lower concentration (20, 40 $\mu\text{g/ml}$) (Fig. 1(5)). Interestingly, CASE potentiated TGF- β_1 -induced activation and expression of pp38 in HepG2 cells similar to HSCs (Fig. 1(6)).

3.2. CASE decreased TGF- β_1 -induced domain-specific phosphorylation of Smad2/3 and nuclear translocation of Smad4 in both HSCs and HepG2 cells

Stimulation of HSCs with exogenous TGF- β_1 resulted in increased expression of phosphorylated pSmad2C, pSmad2L and oncogenic pSmad3L, however prior treatment of HSC cells with CASE (20, 40 and 80 $\mu\text{g/ml}$ respectively) before TGF- β_1 stimulation led to a

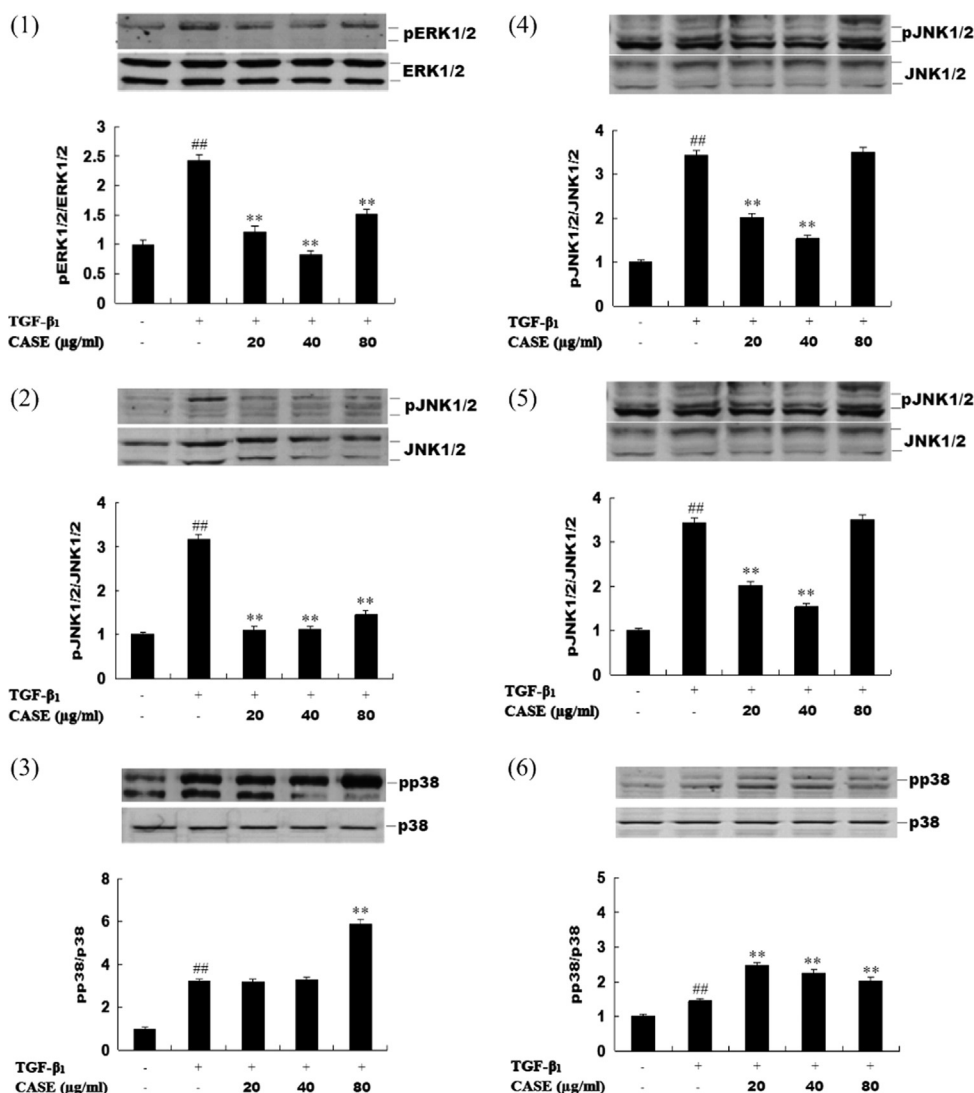


Fig. 1. Effect of CASE on TGF- β_1 -induced activation of pERK, pJNK and pp38 in HSCs (Fig. 1(1), (2) and (3)) and HepG2 Cells (Fig. 1(4), (5) and (6)). The HSCs and/or HepG2 cells were seeded at a density of 1×10^6 cells/25 cm^2 culture flasks then cultured with 10% FBS in 95% air and 5% CO_2 at 37 $^\circ\text{C}$. The cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 $\mu\text{g/ml}$) when the HSCs and/or HepG2 cells attached to 70–80% of monolayers, subsequently cells were treated with TGF- β_1 (9 pmol/L) for 1 h. Total proteins of the cells were extracted by using Western blot and IP cell lysis liquid. Expression of ERK1/2, JNK1/2 and pp38 were monitored by western blot using anti-pERK1/2, anti-pJNK1/2 and anti-pp38 antibodies respectively. Intensities of pERK1/2, pJNK1/2 and pp38 bands were normalized to ERK1/2, JNK1/2 and p38 respectively of the corresponding treatment groups. The data presented are based on at least three independent experiments. (^{##} $P < 0.01$ compared with control group; ^{**} $P < 0.01$ compared with TGF- β_1 group).

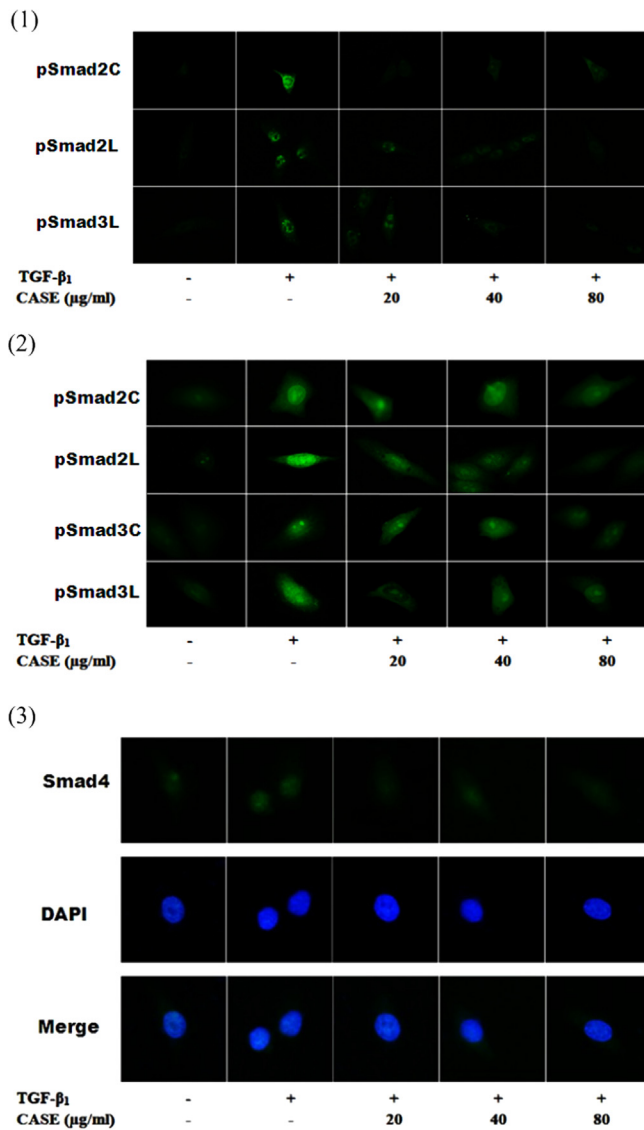


Fig. 2. Effect of CASE on TGF- β_1 -induced domain-specific phosphorylation of Smad2/3, Smad4 expressions and their nuclear translocation in HSCs (Fig. 2(1)) and HepG2 cells (Fig. 2(2) and (3)). HSCs and/or HepG2 cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 μ g/ml) when the cells attached to 30–40% of monolayers in 24-holes culture plate, respectively; subsequently the cells were treated with TGF- β_1 (9 pmol/L) for 1 h. The cells were then fixed, permeabilized, blocked and incubated with each primary antibody, then corresponding fluorescein isothiocyanate (FITC)-Conjugated secondary antibody, ultimately viewed and photographed using a fluorescence microscope. Fig. 2(3)) HepG2 cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 μ g/ml) when the cells attached to 30–40% of monolayers in 24-holes culture plate respectively; subsequently cells were treated with TGF- β_1 (9 pmol/L) for 1 h. The cells were then fixed, permeabilized, blocked and incubated with each primary antibody, then corresponding fluorescein isothiocyanate (FITC)-Conjugated secondary antibody, incubated with 4', 6-diamidino-2-phnylindole (DAPI) for nuclear staining, ultimately viewed and photographed using a fluorescence microscope. C, C-terminal; L, Linker region.

concentration-dependent decrease in the TGF- β_1 -induced phosphorylation of pSmad2C, pSmad2L and oncogenic pSmad3L (Fig. 2(1)). In a similar manner, treatment of HepG2 cells with TGF- β_1 (9 pmol/L) produced increased domain-specific phosphorylation and expression of pSmad2C, pSmad2L, oncogenic pSmad3L and Smad4 (Fig. 2(2) and (3)). But pretreatment of HepG2 cells with CASE (20, 40 and 80 μ g/ml respectively) before TGF- β_1 stimulation, led to the reversal of the above observations. Importantly, CASE reduced nuclear import of Smad4 by enhancing its cytoplasmic retention and also enhanced

the phosphorylation of tumor suppressor pSmad3C, specifically in HepG2 cells (Fig. 2(2) and (3)).

3.3. CASE decreased TGF- β_1 -induced expression of Imp7/8 and their subcellular localization in both HSCs and HepG2 cells

TGF- β_1 stimulated HSCs produced not only increased protein expression of Imp7/8 but also enhanced nuclear relocation of Imp7/8 compared to control (Fig. 3(1) and (2)). But CASE treatment before TGF- β_1 stimulation produced a concentration-dependent decrease in Imp7/8 expression and also reduced significantly their nuclear relocation, especially Imp7 (Fig. 3(5)). Importantly, CASE (40 and 80 μ g/ml) was most effective. Similarly, TGF- β_1 -stimulated HepG2 cells produced increased expression of Imp7/8 and also enhanced nucleo-cytoplasmic distribution. But prior treatment of HepG2 cells with CASE before TGF- β_1 stimulation produced not only reduction in the expressions of Imp7/8 (Fig. 3-(3), (4)) but also reduced their nuclear relocation, particularly Imp7 (Fig. 3(6)).

3.4. CASE repressed TGF- β_1 -induced expression of PAI-1 gene in HSCs and HepG2 cells

TGF- β_1 (9 pmol/L) stimulation of HSCs and HepG2 cells in separate experiments produced increased PAI-1 gene expression compared to control groups, but upon prior CASE treatment before TGF- β_1 stimulation produced a concentration dependent repression of PAI-1 gene expression in the studied cell lines in a reverse manner (Fig. 4(1) and (2)).

3.5. CASE decreased DEN-induced activation of pERK, pJNK and pp38 in rats

DEN time dependently increased the activation and expression of pERK1/2, pJNK1/2 and pp38 in rat livers. However, concurrent treatment of rats with DEN (0.5 ml/100 g bodyweight) in the morning and CASE (60, 120 and 240 mg/kg respectively) in the afternoon first for a period of 12 weeks and subsequently for a period of 16 weeks showed dose dependent reduction in the DEN-induced activation and expression of pERK1/2, pJNK1/2 and pp38 (Fig. 5). With the exception of CASE (60 mg/kg) which produced weak inhibitory effect on DEN-induced activation and expression of pJNK1/2 at the end of the 12 weeks (Fig. 5(2)), CASE (60, 120 and 240 mg/kg respectively) dose dependently decreased DEN-induced activation and expression of pERK, pJNK and pp38 at both 12th and 16th weeks (Fig. 5). CASE (120 mg/kg) was most effective in decreasing DEN-induced activation and expression of pERK1/2 and pJNK1/2 at both the 12th and 16th weeks, while CASE (240 mg/kg) was only effective in decreasing the activation and expression of pp38 at both periods (Fig. 5(3) and (6)).

4. Discussion

We herein present a study further elucidating the molecular mechanisms of CASE as a potential anti-HCC herbal medicine. Essentially, we report that CASE modulates the MAPK-regulated TGF- β /Smad signaling via inhibition of oncogenic MAPK-dependent linker phosphorylation of Smad2/3 in HSCs (A key hepatic cell implicated in liver fibrosis (Puche et al., 2013) and HepG2 cells (A kind of human hepatoma cell line) resulting in down-regulation of PAI-1 gene. Precisely, CASE also inhibited the MAPK pathway in DEN-induced HCC rats, which provided a strong rationale for us to probe further, how the inhibition of the MAPK pathway will affect the overall TGF- β /Smad signaling. The MAPK pathway is one of the important signaling collaborators of TGF- β that enhances overall oncogenic TGF- β /Smad signaling output in cancer (Giehl et al., 2007; Moustakas and Heldin, 2005). Indeed, the cross-

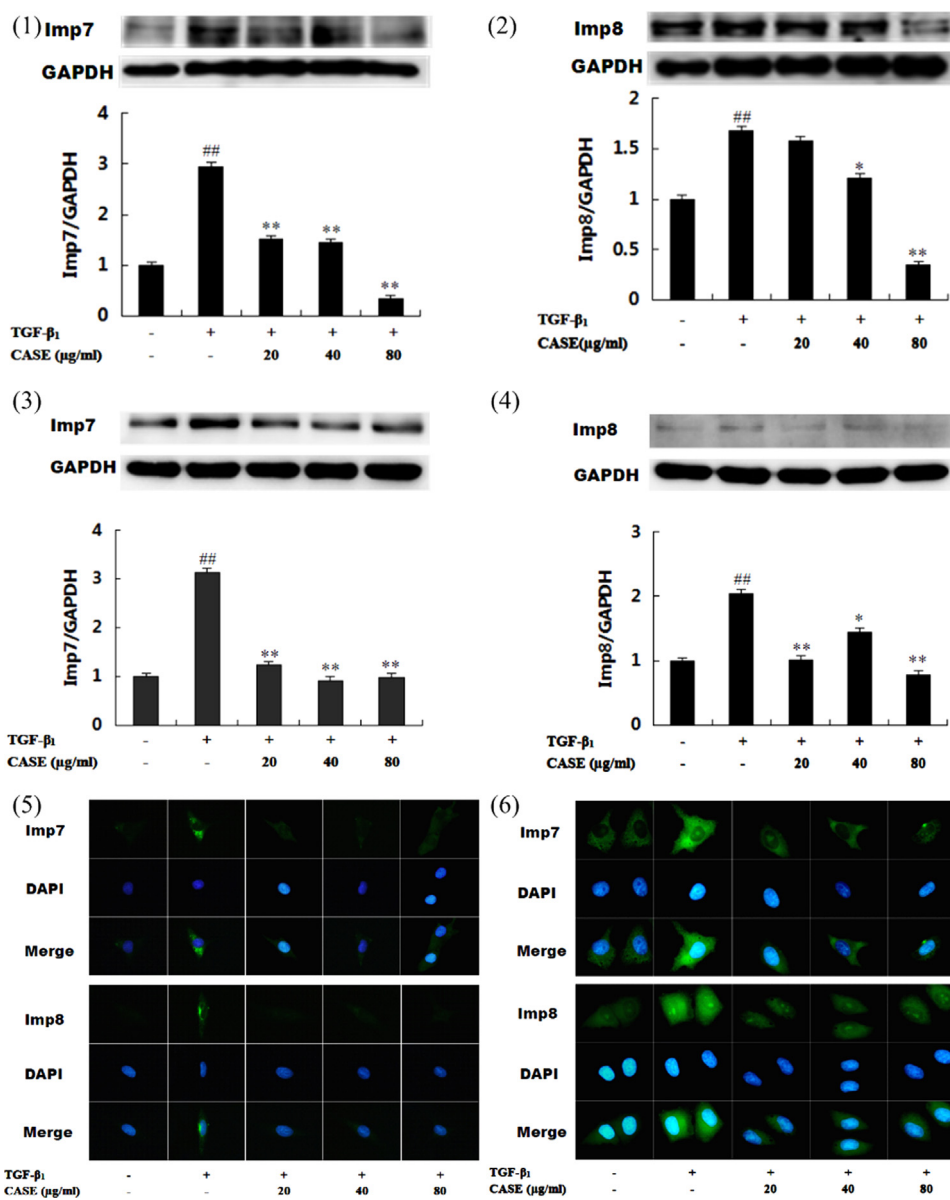


Fig. 3. Effect of CASE on TGF- β_1 -induced expression of Importin7 (Imp7) and Importin8 (Imp8) in HSCs (Fig. 3(1), (2) and (5)) and HepG2 cells (Fig. 3(3), (4) and (6)). The HSCs and HepG2 cells were seeded at a density of 1×10^6 cells/25 cm² culture flasks, then cultured with 10% FBS in 95% air and 5% CO₂ at 37 °C. The cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 μg/ml) when the HSCs and/or HepG2 cells attached to 70–80% of monolayers, subsequently cells were treated with TGF- β_1 (9 pmol/L) for 1 h. Total Imp7/8 proteins of the cells were extracted by using Western blot and IP cell lysis liquid. Expression of Imp7 and Imp8 were monitored by IB using anti-Importin 7 and anti-Importin 8 antibodies respectively. Intensities of Imp7 and Imp8 bands were normalized to glyceraldehyde-3-phosphate-dehydrogenase (GAPDH) of the corresponding treatment groups. The ratio of the Imp7 and Imp8 to GAPDH without exogenous TGF- β_1 was assigned a value of 1. And Fig. 3(5) and (6)) HSC and/or HepG2 cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 μg/ml) when the cells attached to 30–40% of monolayers in 24-holes culture plate, respectively; subsequently cells were treated with TGF- β_1 (9 pmol/L) for 1 h. The cells were then fixed, permeabilized, blocked and incubated with each primary antibody, then corresponding fluorescein isothiocyanate (FITC)-conjugated secondary antibody, incubated with 4, 6-diamidino-2-phrynolindole (DAPI) for nuclear staining, ultimately viewed and photographed using a fluorescence microscope. The data presented are based on at least three independent experiments. (* $P < 0.05$, ** $P < 0.01$ compared with control group; * $P < 0.05$, ** $P < 0.01$ compared with TGF- β_1 group).

signaling between the MAPK pathway and the TGF- β /Smad signaling is now a common phenomenon reported in many cancer subtypes. For instance, the MAPKs have been shown to be activated by TGF- β in various cancer subtypes including HCC (Zhang, 2009) and the TGF- β -induced activation of the MAPKs have been shown to increase MAPK-dependent linker phosphorylation of Smad2/3 (Furukawa et al., 2003; Yoshida et al., 2005) which incidentally promotes oncogenesis (Nagata et al., 2009). Notably, activation of ERK1/2 and JNK was shown to be crucial for TGF- β -induced Smad4-independent signaling (Giehl et al., 2007) which means that the MAPK pathway can significantly promote TGF- β signaling even in the absence of Smad4 (A common Smad protein which mediates canonical TGF- β signaling) via the linker phosphorylation of

Smad2/3 (Non-canonical TGF- β signaling). Targeted inhibition of the MAPK pathway, specifically inhibition of MAPK-dependent linker phosphorylation of the R-Smads may significantly abrogate TGF- β /Smad signaling in cancer. Quiet recently, by using MAPK-Specific Inhibitors (PD98059 [ERK-Specific inhibitor], SP600125 [JNK-Specific inhibitor] and SB203580 [p38-Specific inhibitor]), our study group reported on effective modulation of the MAPK-regulated TGF- β /Smad signaling in HCC leading to decreased PAI-1 gene expression and near blockade of HCC phenotypes (Cell proliferation, migration and invasion) (Boye et al., 2015). In this study, CASE inhibited hitherto increased TGF- β_1 -induced activation of pERK and pJNK in HSC and HepG2 cells and these results were consistent in both HSC and HepG2 cells. This shows that CASE

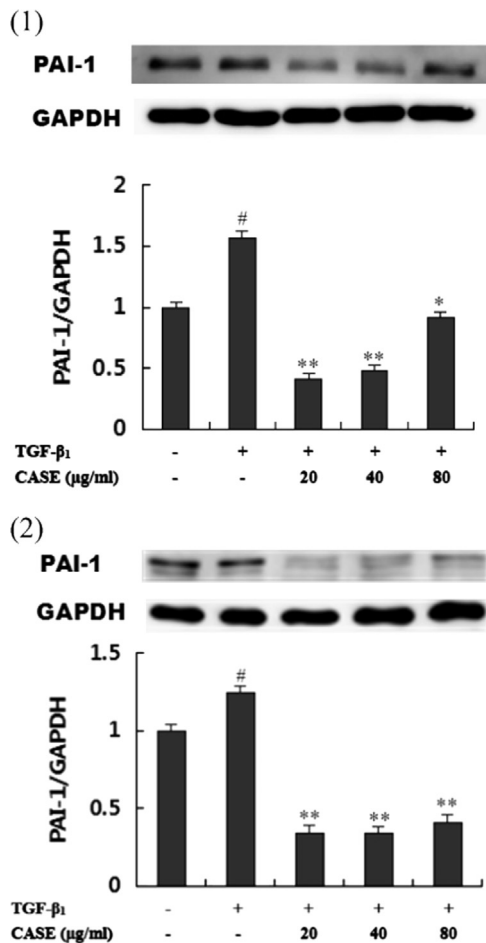


Fig. 4. Effect of CASE on TGF- β_1 -induced expression of plasminogen activator inhibitor 1 (PAI-1) in HSCs (Fig. 4(1)) and HepG2 cells (Fig. 4(2)). The HSCs and/or HepG2 cells were seeded at a density of 1×10^6 cells/25 cm² culture flasks, then cultured with 10% FBS in 95% air and 5% CO₂ at 37 °C. The cells were starved for 24 h in serum-free medium and in the absence or presence of CASE (20, 40 or 80 $\mu\text{g/ml}$) when the HSCs and/or HepG2 cells attached to 70–80% of monolayers, subsequently cells were treated with TGF- β_1 (9 pmol/L) for 1 h. Total proteins of the cells were extracted by using Western blot and IP cell lysis liquid. Expression of PAI-1 gene was monitored by western blot using anti-PAI-1 Ab. Intensities of PAI-1 bands were normalized to glyceraldehyde-3-phosphate-dehydrogenase (GAPDH) of the corresponding treatment groups. The ratio of the PAI-1 gene to GAPDH without exogenous TGF- β_1 was assigned a value of 1. The data presented are based on at least three independent experiments. (# P < 0.05 compared with control group; * P < 0.05, ** P < 0.01 compared with TGF- β_1 group).

in vitro target ERK/JNK-dependent MAPK signaling in the studied cell lines, however, CASE increased pp38 in both cell lines even more than the model group, which perhaps indicates that CASE variably modulates the MAPK pathway *in vitro*. Activation of the MAPKs by TGF- β and subsequent MAPK-dependent linker phosphorylation of Smad2/3 enhances nuclear import of pSmad2/3 to promote TGF- β signaling. In view of this we sought to find out how CASE-dependent inhibition of the MAPKs in HSC and HepG2 cells will affect the down and upstream mediators of the TGF- β /Smad signaling, specifically, we monitored domain-specific phosphorylation of Smad2/3, expression of Smad4 and Imp7/8 and their nuclear import.

Briefly, TGF- β signaling begins with ligand activation of constituted TGF- β receptor type 2 (T β RII), which then trans-phosphorylate TGF- β receptor type 1 (T β RI). The phosphorylated T β RI in turn phosphorylate receptor-dependent Smad proteins (Smad2 and Smad3) at the C-terminal to facilitate their oligomerization with Smad4 to form a complex (Smad2/3/4 complex) which preferentially relocates into the nucleus (Matsuzaki, 2012). In the nucleus, pSmad2/3 directly partakes in the execution of TGF- β -mediated transcription of target genes

through association with transcriptional effectors (Promoters and repressors) (Matsuzaki, 2012; Murata et al., 2014). Thus, the Smads act as both signaling mediators (Messengers) and transcriptional effectors to determine the fate of TGF- β -specific target gene transcription of which PAI-1 gene is integral. Critical to both canonical and non-canonical TGF- β signaling is the nuclear import of pSmad2/3 as a complex with Smad4 (Smad2/3/4 complex) or through Smad4-independent mechanism via MAPK-dependent linker phosphorylation of Smad2/3. Essentially, any of the two import mechanisms may involve Imp7/8. For instance, (Chen and Xu, 2011) have shown that importin proteins (Imp7/8) are indispensable for nuclear import of pSmad2/3 and Smad4. It was also shown that nuclear import of pSmad2/3 is a critical step in TGF- β signaling transduction (Xu et al., 2007). Further, it was shown that knockdown of importin 7/8 markedly impaired nuclear import of pSmad2/3 in response to TGF- β -dependent gene regulation which is critical for embryonic development and cellular homeostasis but detrimental in dysregulated TGF- β signaling in cancer (Xu et al., 2007). Also, (Xu et al., 2007) have intimated that since Imp7/8 is critical for nuclear import of pSmad2/3 it is highly probable that pharmacological inhibition of this nuclear import factor may disrupt dysregulated TGF- β /Smad signaling in cancer. Accordingly, TGF- β_1 stimulation of HSCs and HepG2 cells produced increased linker phosphorylation of Smad2/3, increased expression of Smad4 and Imp7/8 and their nuclear import and this correlated with increased PAI-1 gene expression in both cell lines (Fig. 4(1) and (2)). Interestingly, CASE-dependent inhibition of pERK and pJNK in both cell lines correlated with CASE-dependent inhibition of linker phosphorylation of Smad2/3, decreased expression of Smad4 and Imp7/8 and their nuclear import. For example, in HepG2 cells CASE significantly decreased nuclear import of Smad4 by enhancing its cytoplasmic retention (Fig. 2(2) and (3)). Also, CASE treatment inhibited Imp7/8 expression compared with model (TGF- β_1 -stimulated HSCs and HepG2 cells) and control groups. Precisely, CASE decreased both cytoplasmic and nuclear expression of Imp7 but only decreased cytoplasmic expression of Imp8 compared to model and control groups in both HSC and HepG2 cells and these effects correlated with decreased pSmad2/3 and Smad4 expressions (Fig. 2 (1), (2) and (3)) which suggest that CASE-dependent inhibition of pSmad2/3 and Smad4 nuclear import may be mediated through Imp7 inhibition. Correspondingly, CASE-dependent decrease in nuclear import of pSmad2/3, Smad4 and Imp7 led to significant decrease in PAI-1 gene expression.

Severally, PAI-1 gene has been reported as one of the important target genes of TGF- β signaling which has been shown to account for most of the pathological roles of TGF- β signaling in cancer (Derynck and Zhang, 2003; Tahashi et al., 2002). Functionally, PAI-1 gene acts as the main inhibitor of the urokinase-type plasminogen activator system and it stimulates cell migration and invasion by disrupting cellular adhesion and enhancing basement membrane degradation (Gutierrez et al., 2000). Indeed, overexpression of PAI-1 gene has been implicated in multiple forms of fibrosis including liver fibrosis (Gramling and Church, 2010), a key risk factor of HCC. In our study, both TGF- β_1 -stimulated HSC and HepG2 cells showed increased expression of PAI-1 gene. This observation is consistent with our previous reports regarding the ability of CASE to modulate TGF- β /Smad signaling leading to decrease in PAI-1 gene expression in myofibroblast (Yang et al., 2008), keloid fibroblast (He et al., 2012), HepG2 cells (Hu et al., 2014; Liu et al., 2010), and DEN-induced HCC in rats (Rui et al., 2014). Clearly, from the present results CASE selectively modulate ERK/JNK-dependent linker phosphorylation of Smad2/3 *in vitro*, in part this observation confirms our earlier report concerning JNK/MAPK-regulated TGF- β /Smad signaling in myofibroblast (Yang et al., 2008) where CASE in a concentration dependent manner inhibited JNK and JNK-dependent linker phosphorylation of Smad2/3.

To further confirm or otherwise the ability of CASE to modulate the MAPK pathway, we studied the effect of CASE on DEN-induced

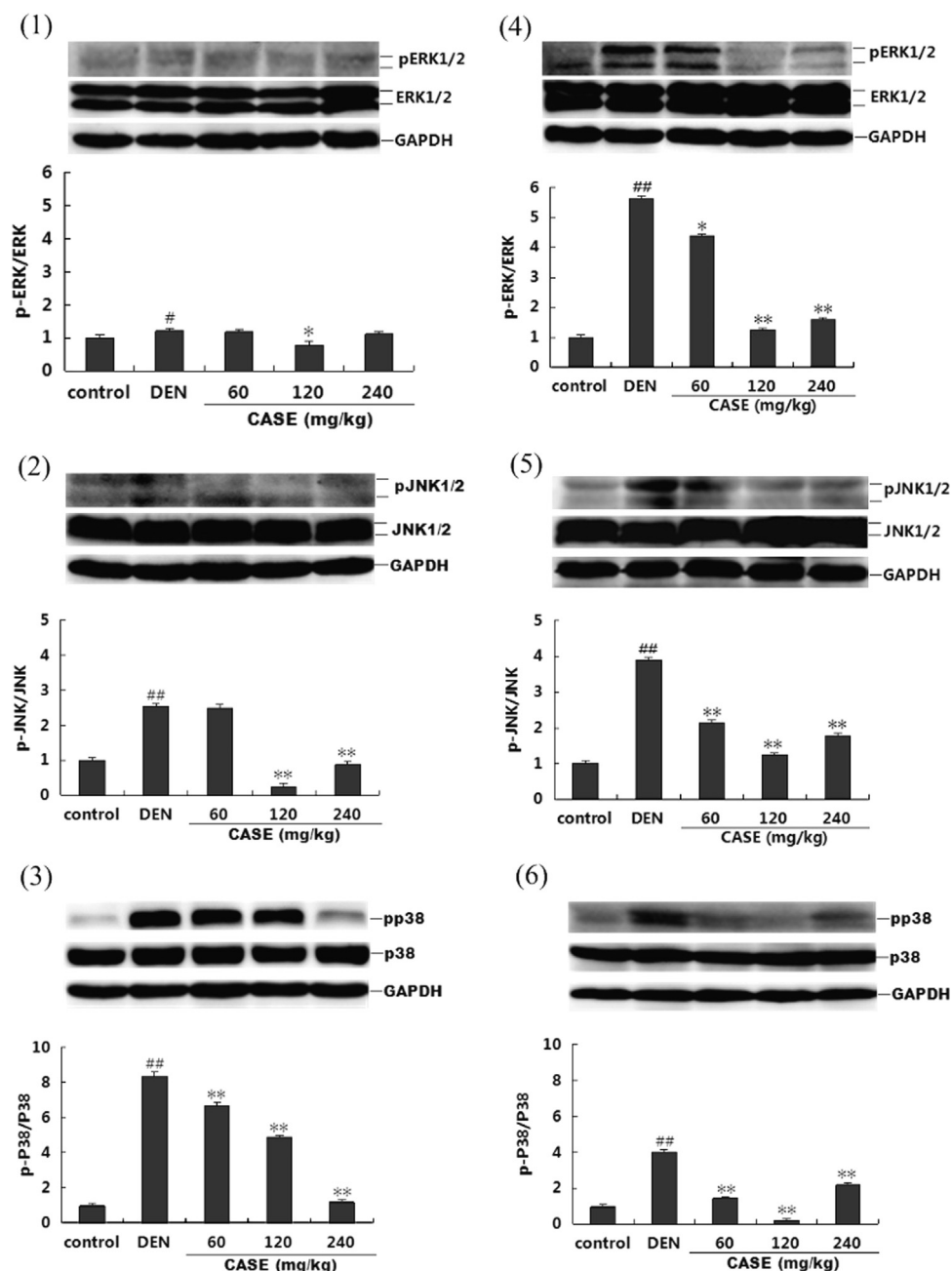


Fig. 5. Effect of CASE on diethyl nitrosamine (DEN)-induced activation of pERK, pJNK and pp38 in rats. The proteins were extracted from frozen liver tissues from 12th week (Fig. 5(1), (2) and (3)) and 16th week (Fig. 5(4), (5) and (6)) DEN treated rats. Phosphorylated (p) ERK1/2, pJNK1/2, pp38 and ERK, JNK, p38 were analyzed by western blot using anti-pERK1/2, pJNK1/2, pp38 and ERK1/2, JNK1/2, p38 Abs and anti-glyceraldehyde phosphate dehydrogenase (GAPDH) Ab. Intensities of pERK1/2, pJNK1/2, pp38 bands were normalized to those of ERK1/2, JNK1/2, p38 of the corresponding treatment groups. The ratio of the pERK1/2, pJNK1/2 and pp38 proteins to ERK1/2, JNK1/2, and p38 respectively of normal group was assigned a value of 1. Each value represents mean $\pm$ SD, $n=3$. $^{\#}P < 0.05$, $^{##}P < 0.01$ compared with control group; $^*P < 0.05$, $^{**}P < 0.01$ compared with DEN group.

activation of the MAPK pathway. From our *in vivo* study, treatment of rats with DEN for 12 weeks led to increased activation and expression of pERK, pJNK and pp38; but extending DEN treatment period to 16 weeks even saw a further increase in the average expression of pERK, pJNK and pp38 (Fig. 5(4), (5) and (6)), indicating a direct causal relationship between duration of DEN treatment and MAPK activation. These results indicate that increase in MAPK activation and expression may correlate with HCC progression. But concomitant treatment of rats with DEN in the morning and CASE in the afternoon reversed the above relationship, for instance at 12 weeks not only did CASE (120 or 240 mg/kg) decreased the expression of pERK, pJNK and pp38

compared to the model group, but also decreased the average expression level of each MAPK even more than the average for the control group (Fig. 5(1), (2) and (3)) showing that CASE potently modulates the MAPK pathway and that the MAPK pathway could be yet another important target of CASE. At 16 weeks, though CASE decreased the expression of the MAPKs compared to the model group, it was however not different from that of the control group (Fig. 5(4), (5) and (6)) indicating that MAPK activation and expression were indeed related to duration of DEN treatment. The *in vivo* inhibition of pp38 by CASE (Fig. 5(3) and (6)) contradicts our *in vitro* results regarding pp38. But this observation probably suggest that CASE

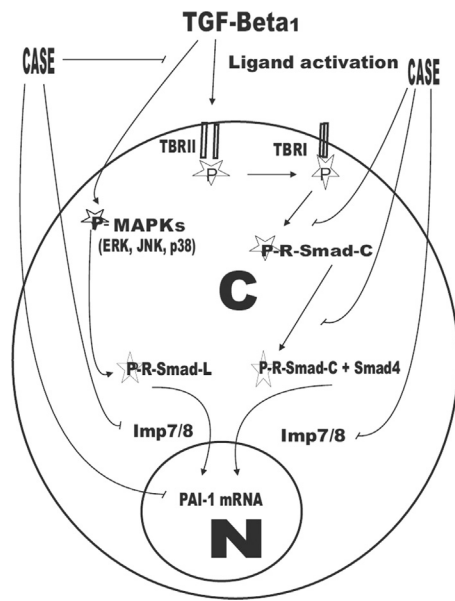


Fig. 6. Fig. 6 shows an illustrative summary of the multi-target mechanism by which CASE modulates the MAPK-regulated TGF- β /Smad signaling in hepatocellular carcinoma (HCC). CASE blocks TGF- β -induced activation of ERK, JNK, and p38 to abrogate MAPK-dependent linker phosphorylation of Smad2/3 and their subsequent nuclear translocation. Along the canonical loop, CASE inhibits Smad4 expression, formation of Smad2/3/4 and their nuclear translocation via inhibition of Imp7/8. CASE-dependent disruption of nuclear import of Smad2/3 and Smad4 leads to down-regulation of PAI-1 gene expression. Cytoplasm (C), C-terminal phosphorylated Smad (pSmadC), Linker phosphorylated Smad (pSmadL), Nucleus (N), Phosphorylated Smad (pSmad), Receptor mediated Smad (R-Smad), Transforming growth factor receptor type 1 and 2 (T β RI and T β RII)

modulate pERK, pJNK and pp38 *in vivo* by the same mechanism which may involve some unknown factors in the tumor microenvironment, since the tumor microenvironment was the only major factor absent in the *in vitro* study or it may be possible that while CASE employs the same mechanism to modulate pERK and pJNK both in *in vitro* and *in vivo*, it, however, employs different mechanisms to modulate pp38 in *in vitro* and *in vivo* settings. Indeed, these results further show that CASE does not only target the upstream and downstream mediators (TGF- β ligand, T β RI, T β RII, Smad2C/L, Smad3L, Smad4, and PAI-1) of TGF- β /Smad signaling as previously reported (He et al., 2012; Hu et al., 2014; Liu et al., 2010; Rui et al., 2014; Wu et al., 2014; Yang et al., 2008) but also targets the MAPK pathway. Fundamentally, our *in vivo* results agree in part with (Nagata et al., 2009), with regard to JNK activation by DEN treatment, however, our results concerning pERK and pp38 are inconsistent with their finding. For instance, Nagata et al. (2009) had shown that after 86-days DEN treatment of rats, only pJNK was activated while pERK and pp38 remained inactive (Un-phosphorylated), but in our study we found that increasing DEN treatment to 12 or 16 weeks produced significant activation of all the MAPKs (pERK, pJNK and pp38). We report that activation of the MAPKs by DEN treatment is time-dependent and may relate directly to HCC progression.

Put together, the present results provide a strong rationale for further elucidation of the MAPK pathway as a key target for therapy against HCC. Essentially, CASE has shown a multi-target potential by modulating both the MAPK and TGF- β /Smad signaling pathways (Fig. 6) in a manner that highlights its potential drug candidature. However, our results raise some further questions, for example, which specific subcellular location (Cytoplasm or nucleus) does CASE modulate the MAPK pathway and how does it affect HCC phenotypic manifestations such as cell proliferation, dysregulated apoptosis, cell migration and invasion? These questions are informed by the fact that spatio-temporal regulation of

the MAPK pathway is crucial for many pathways implicated in HCC including multi-functional TGF- β /Smad signaling.

5. Conclusion

CASE blocked MAPK activation and MAPK-dependent linker phosphorylation of Smad2/3, inhibited Smad4 and Imp7/8, blocked nuclear import of Smad2/3/4 via Imp7 modulation and down-regulated PAI-1 gene expression, and these results do not only demonstrate the potential of the MAPK-regulated TGF- β /Smad pathway as an important target for therapy against HCC but also highlights the potential drug candidature of CASE and confirmation of its ethnobotanical usage.

Conflicts of interest

No potential conflicts of interest.

Acknowledgments

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Glossary

- Phosphorylation: Addition of a phosphorus atom to a protein or a molecule.;
- Smad proteins: A class of proteins that mediate transforming growth factor beta (TGF- β) signaling transduction.;
- Phosphorylated Smad: A Smad protein that has received a phosphorus atom from a phosphorus carrier molecule.;
- Linker phosphorylated Smad: A Smad protein phosphorylated at a region in between its N-terminal and C-terminal.;
- Canonical TGF- β Signaling: It is Smad-mediated TGF- β signaling transduction involving Smad2, Smad3, and Smad4 complex formation leading to their nuclear translocation and subsequent transcription of TGF- β -target mRNAs.;
- Importin protein 7 and 8 (Imp7/8): They are proteins mostly in the cytoplasm that facilitate nuclear entry of other proteins.;
- Mitogen activated protein kinase (MAPK): They are kinase proteins that facilitate transduction of extracellular signals into the nucleus.;
- Non-canonical TGF- β signaling: All TGF- β signaling transduction that does not involve Smad4 or non-Smad TGF- β signaling..